Gut Microbial Diversity in Amyotrophic Lateral Sclerosis: A Reanalysis of the Hertzberg Cohort

Project Report - BIOL 3601, Spring 2026

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Abstract

Amyotrophic lateral sclerosis (ALS) is a fatal motor neuron disease, and most cases have no known cause. One area that has received recent attention is the gut microbiome, which can influence the brain through the gut-brain axis. To test whether the gut bacterial community of ALS patients is different from that of healthy people who live with them, I reanalyzed 16S ribosomal RNA sequencing data for nine ALS patients and their nine spouses (Hertzberg et al. 2021) from NCBI BioProject PRJNA566436. The reads were processed with DADA2 (Callahan et al. 2016) against the SILVA 138 reference database (Quast et al. 2013), which produced 275 amplicon sequence variants (ASVs) for the 18 samples. After a prevalence filter and conversion to relative abundances, I computed the Shannon index for alpha diversity and used a paired Wilcoxon signed-rank test to compare the two groups within household pairs. For beta diversity, I computed Bray-Curtis dissimilarities, visualized them with Principal Coordinates Analysis (PCoA), and tested group separation with PERMANOVA. A Mann-Whitney scan with Benjamini-Hochberg correction was used to identify the driver taxa. The Shannon result was the opposite of what I had predicted: ALS patients had higher diversity than their spouses (paired Wilcoxon $p = 0.004$). The overall composition was also significantly different between the two groups (PERMANOVA pseudo-F = 1.7, p approximately 0.02), although the effect was small. At the genus level, several taxa showed consistent enrichment in one of the two groups. The literature on ALS alpha diversity is not consistent, and these results add another data point to that picture.

1 Introduction

Amyotrophic lateral sclerosis (ALS) is a fatal disease of the motor neurons in the brain and spinal cord. As the motor neurons die, patients gradually lose voluntary muscle control. They can no longer walk, then they have difficulty swallowing and speaking, and eventually they cannot breathe without assistance (Brown and Al-Chalabi 2017). About 90 percent of cases are sporadic, which means the patient has no family history of ALS. Several risk genes have been identified, but no single cause has been confirmed for sporadic ALS after many decades of research. Because no clear cause has been found in the central nervous system itself, researchers have begun to investigate factors outside it, including the gut microbiome.

The gut microbiome is the community of trillions of bacteria that live in the digestive tract. It sends signals to the brain through three main routes: the vagus nerve, the immune system, and bacterial metabolites that travel through the bloodstream. This communication network is called the gut-brain axis (Cryan et al. 2019). When the bacterial community shifts away from a healthy state, a condition called dysbiosis, signals along all three routes can be disturbed. Dysbiosis has been reported in many disorders, including inflammatory bowel disease, depression, Parkinson's disease, and Alzheimer's disease. The broad pattern across these conditions tends to be similar: a loss of certain beneficial bacteria, often short-chain fatty acid producers, together with an increase in less common taxa associated with inflammation. The specific shifts differ from disease to disease, but the recurring presence of a microbiome signal in conditions with no obvious gastrointestinal component is what has made the gut-brain axis a serious research direction. The connection is particularly relevant for ALS because it suggests that part of the disease mechanism may be peripheral and therefore potentially modifiable. The proposed mechanisms work in both directions: changes in the brain can affect gut motility and immune function, and changes in the gut can affect brain function through any of the three channels of the axis.

The most direct evidence for a connection between the gut microbiome and ALS comes from a 2019 study by Blacher and colleagues, who used the SOD1 mouse model (Blacher et al. 2019). Depleting the gut microbiota with antibiotics made the disease worse, while supplementing the mice with *Akkermansia muciniphila* slowed its progression. The protective effect was traced to a bacterial metabolite called nicotinamide. The same study also reported lower cerebrospinal-fluid nicotinamide in a small group of human ALS patients. Several 16S surveys of human ALS cohorts have been published since then, reporting differences in alpha diversity and in the relative abundance of major bacterial phyla, but the direction and the magnitude of the reported effects are not consistent across studies, and no clear signature has been established.

For this project I reanalyzed one of these human cohorts to test whether the gut bacterial community in ALS patients differs from that of healthy spouse controls, both in within-sample diversity (alpha) and in between-sample composition (beta). Before examining the data, I expected ALS patients to show lower Shannon alpha diversity than their spouses and to separate visibly from them in a Bray-Curtis ordination. The spouse pairing is the main reason this cohort is informative. Spouses share diet, household, and many environmental exposures, so comparing within a pair removes much of the variation that confounds case-control comparisons drawn from unrelated individuals.

2 Workflow and Methods

2.1 Data Acquisition

The raw data come from the cohort published by Hertzberg and colleagues (Hertzberg et al. 2021), deposited at NCBI Sequence Read Archive (SRA) under BioProject PRJNA566436. The cohort consists of 18 stool samples from nine clinically confirmed ALS patients and the nine spouses they live with. Each ALS patient and the matched spouse form one household pair, and the pairing was preserved throughout the analysis. The household pair identifiers in the metadata run from 1 to 10 with pair 4 excluded upstream, giving nine pairs in total. Sequencing was performed on the Illumina MiSeq platform, targeting the V4 hypervariable region of the 16S ribosomal RNA gene with the standard 515F to 806R primers, producing 250 base-pair paired-end reads. The raw FASTQ files and the quality-filtered FASTQs used as input to the denoising step were retrieved through the published reanalysis repository.

2.2 Preprocessing and Quality Control

Sequence processing followed the standard DADA2 workflow (Callahan et al. 2016). Reads were quality-trimmed, dereplicated, and passed through the DADA2 error model, which infers exact amplicon sequence variants (ASVs) at single-nucleotide resolution rather than clustering reads into approximate operational taxonomic units. Forward and reverse reads were merged, and chimeric sequences generated by PCR artifacts were removed with the DADA2 chimera filter. The chimera-cleaned sequence table contained 275 ASVs across the 18 samples. Taxonomy was assigned to each ASV against the SILVA 138 reference database (Quast et al. 2013) using the naive Bayes classifier included in DADA2. The classifier assigns labels from Kingdom down to Species where the data support the assignment, and leaves the lower ranks empty where they cannot be assigned with confidence.

A prevalence filter was applied before any of the downstream statistics. ASV tables are very sparse, and most ASVs are detected in only one or two samples, where they contribute primarily noise. I removed any ASV that was present in fewer than 5 percent of the samples, with a minimum of two samples for this small cohort. After filtering, 121 ASVs remained, and approximately 75 percent of the entries in the table were still zeros. Each column was then divided by its sum so that the counts became per-sample relative abundances that sum to 1. Relative abundances, rather than raw counts, were used for all of the diversity and ordination calculations because the samples were sequenced to different depths and the raw counts are therefore not directly comparable across samples.

2.3 Statistical and Computational Analysis

Alpha diversity was summarized with the Shannon index (Shannon 1948), computed on the filtered ASV table. Because each ALS patient is matched with their own spouse, the two groups were compared using a paired Wilcoxon signed-rank test on the per-pair Shannon difference rather than a standard two-sample test. The pairing absorbs much of the variation due to diet and household environment, which would otherwise inflate the within-group variance and reduce the power of the test. Beta diversity was computed using the Bray-Curtis dissimilarity (Bray and Curtis 1957) on the per-sample relative abundances. The Bray-Curtis distance matrix was then projected into two dimensions using Principal Coordinates Analysis (Gower 1966) to visualize the data. Group separation in the full distance matrix was tested with PERMANOVA (Anderson 2001), using 999 permutations of the group labels and a fixed random seed so that the result can be reproduced.

To identify the genera that might be driving group separation, the ASV-level relative abundances were aggregated to the genus level using the Genus column of the taxonomy table, and ASVs without a genus assignment were excluded. For each genus, a two-sided Mann-Whitney U test was performed comparing ALS and Control samples. To control the false discovery rate across many simultaneous tests, the raw p-values were adjusted using the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995) and the genera were ranked by adjusted p-value.

2.4 Software and Libraries

The upstream DADA2 pipeline was run in R (version 4.4) with the dada2 package (version 1.32). All downstream analysis was performed in Python (version 3.12) within a dedicated conda environment, using pandas and numpy for table manipulation, scipy for the Wilcoxon and Mann-Whitney tests, scikit-bio for the Shannon index, Bray-Curtis distances, PCoA, and PERMANOVA, statsmodels for the Benjamini-Hochberg correction, and matplotlib together with seaborn for the figures. The exact version pins are stored in the project repository so that the analysis can be reproduced.

3 Results

The prevalence filter reduced the ASV table from 275 to 121 ASVs, while the table remained dominated by zeros (approximately 75 percent sparsity). This reduction is typical for 16S data at this scale and removes the singletons that would otherwise dominate the distance calculations. The filtered counts were converted to per-sample relative abundances and used for every subsequent step of the analysis. The community composition at the phylum level is shown in Figure 1 and provides a baseline view of the data before any group-level statistics are computed. Across the 18 samples the two dominant phyla were Bacteroidota (mean relative abundance approximately 55 percent) and Firmicutes (approximately 34 percent), together accounting for close to 90 percent of each sample, and no obvious phylum-level shift was visible between the ALS and Control halves of the figure.

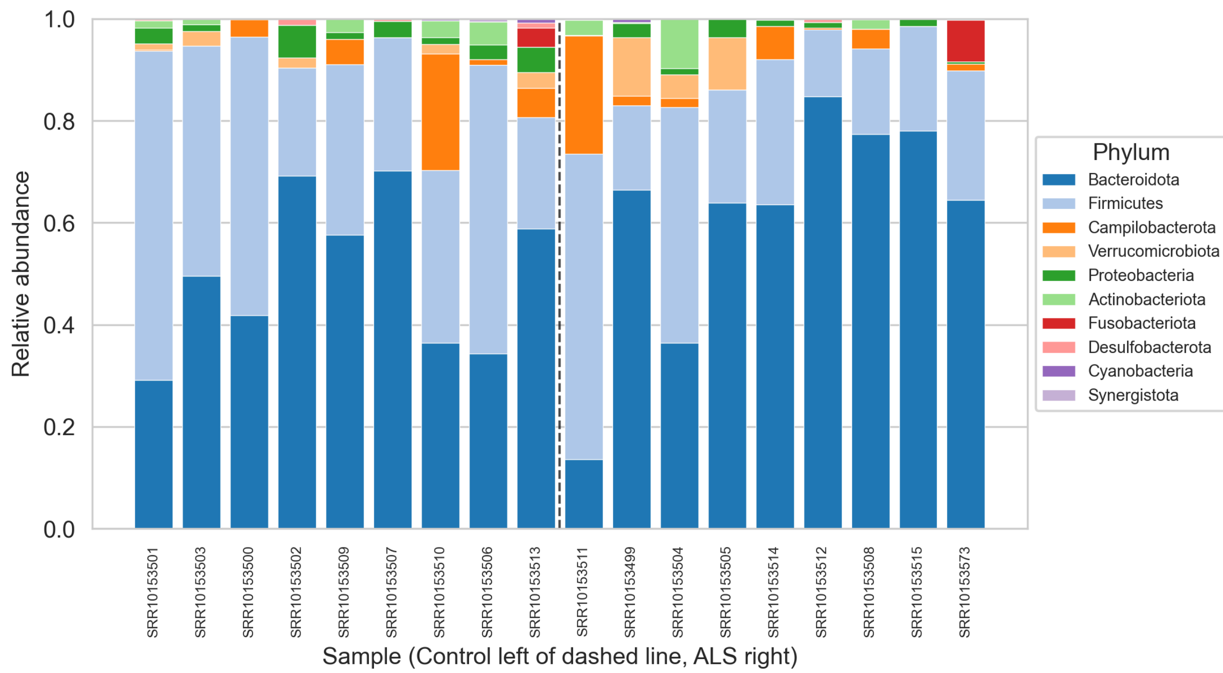

Figure 1: Phylum-level relative abundance for each sample. Samples are ordered by group, with Control on the left of the dashed line and ALS on the right. Phyla are stacked from most to least abundant overall. The two groups have visually similar compositions at the phylum level, with no dominant taxon disappearing in one group.

Alpha diversity differed between the two groups, but in the opposite direction from what I had predicted. The median Shannon index was higher in the ALS group than in the spouse group, and the paired Wilcoxon signed-rank test on the nine per-pair differences gave a p-value of 0.004 (Figure 2). The effect was consistent across the cohort: as Figure 3 shows, in nearly every household pair the ALS patient had a higher Shannon value than their spouse, which is the reason the paired test reaches significance even with only nine pairs.

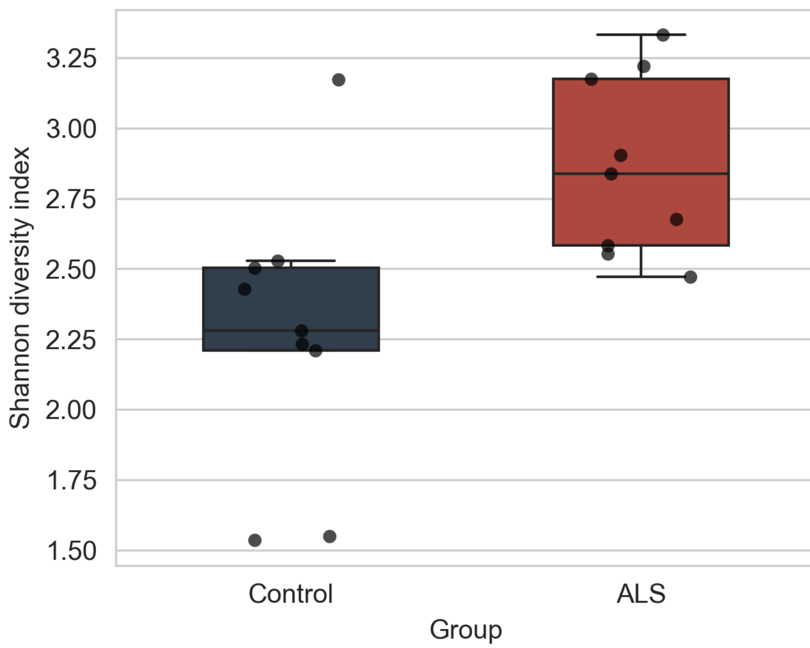

Figure 2: Shannon alpha diversity by group. Each black dot is one sample; boxes show the interquartile range with the median as the horizontal line. ALS patients (red) had higher Shannon diversity than their spouse controls (blue) in nearly every household pair (paired Wilcoxon signed-rank test, $p = 0.004$).

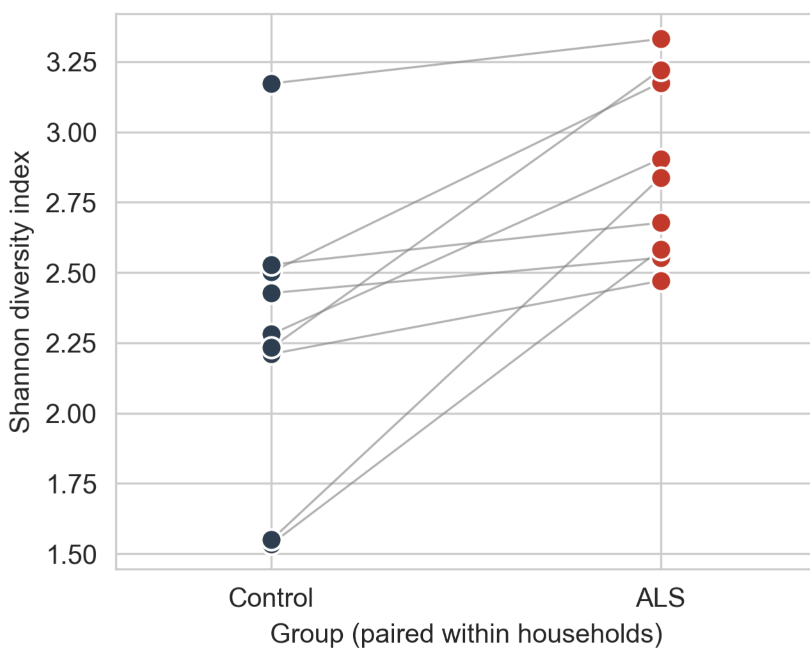

Figure 3: Shannon alpha diversity within each household pair. Each grey line connects the Control sample (left) to the ALS sample (right) of one household. In nearly every pair the line slopes upward, indicating higher Shannon diversity in the ALS patient than in the cohabiting spouse.

Beta diversity produced a weaker but still statistically significant result. The PCoA plot (Figure 4) does not show two clearly separated clusters: the ALS and Control points overlap substantially in the first two coordinates. PERMANOVA on the full Bray-Curtis distance matrix nevertheless detected a group effect (pseudo-F = 1.7, p approximately 0.02 with 999 permutations). The within-group distances are on average smaller than the between-group distances, but the difference is small enough that the visual separation in the plot is not pronounced.

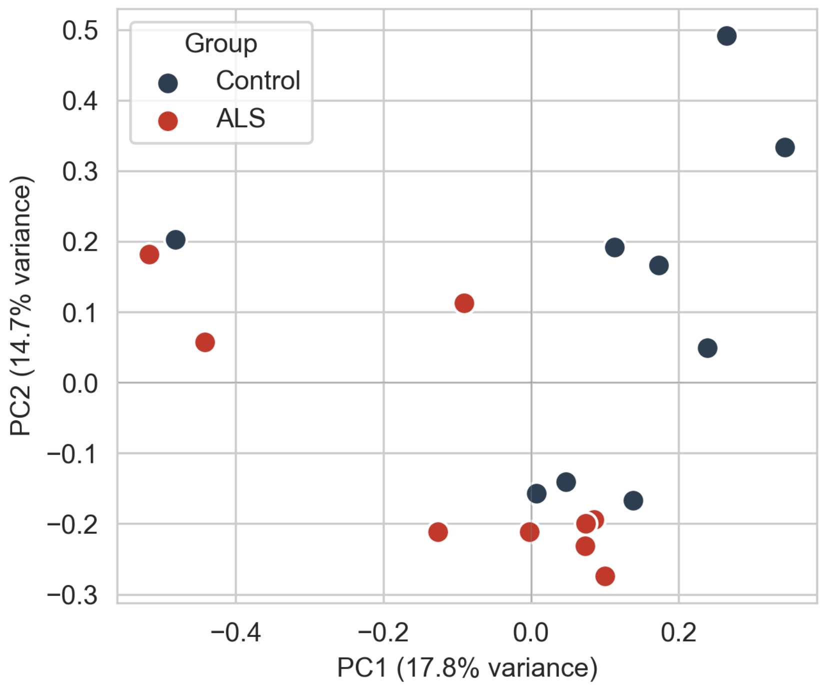

Figure 4: Principal Coordinates Analysis of the Bray-Curtis distances. Each point is one sample, colored red for ALS and blue for Control. The first two principal coordinates explain the variance fraction shown on the axis labels. The two groups overlap visually, but PERMANOVA on the full distance matrix detected significant separation (pseudo-F = 1.7, p approximately 0.02).

At the genus level, the Mann-Whitney scan with Benjamini-Hochberg correction identified a small number of genera that differed in mean relative abundance between the two groups. The full landscape is shown in the volcano plot (Figure 5), and the top ten genera ranked by adjusted p-value are shown in Figure 6. Most of the top hits were enriched in the ALS group, which is consistent with the alpha-diversity result, although the absolute shifts in relative abundance are small. These top genera should be interpreted as leads for further investigation rather than as confirmed biomarkers. With only nine samples per group, each per-genus test is statistically underpowered, and the FDR correction controls only the expected fraction of false positives among the called genera, not the truth of any individual hit. A summary of all main quantitative results is provided in Table 1.

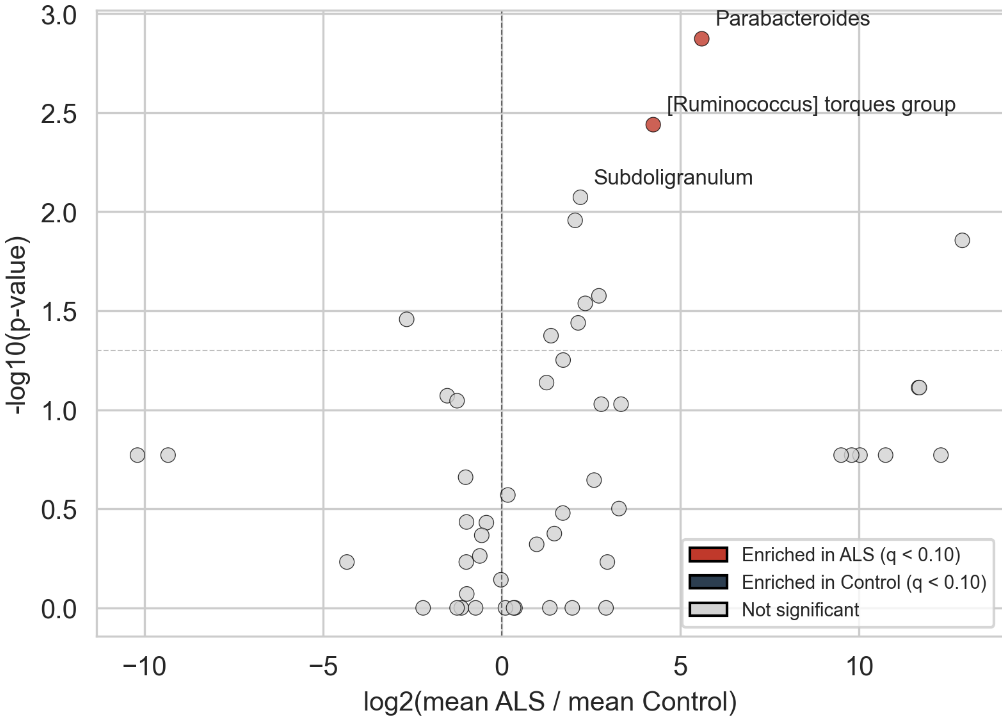

Figure 5: Volcano plot of the per-genus Mann-Whitney U tests. Each point is one genus. The x-axis is the log2 ratio of mean relative abundance in ALS over Control, and the y-axis is the negative log10 of the raw p-value. Genera passing the FDR threshold of $q < 0.10$ are colored by direction of change (red for ALS-enriched, blue for Control-enriched); non-significant genera are grey. The horizontal dashed line marks the nominal $p = 0.05$. The three genera with the smallest adjusted p-values are labeled.

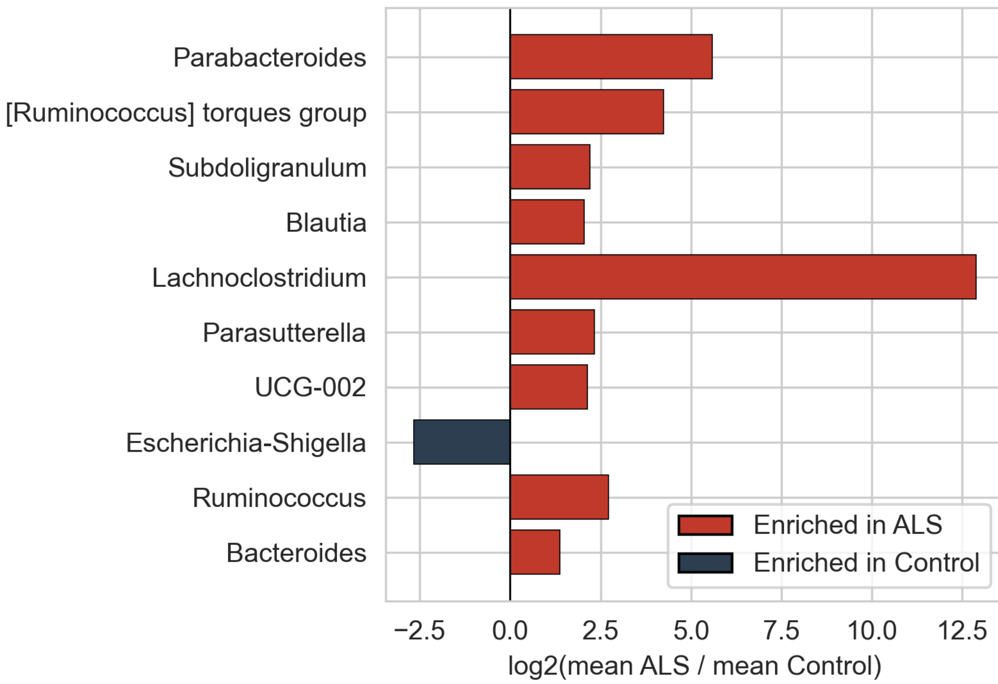

Figure 6: Top 10 bacterial genera ranked by Benjamini-Hochberg adjusted p-value from the per-genus Mann-Whitney U test. Bars to the right of zero indicate enrichment in ALS samples (red); bars to the left indicate enrichment in spouse controls (blue). The x-axis is the log2 ratio of mean relative abundances in ALS vs. Control.

Table 1: Summary of the main quantitative results.

Quantity	Value
ASVs after DADA2 chimera removal	275
ASVs retained after the 5 percent prevalence filter	121
Sparsity after filtering	75 percent
Shannon index, median (ALS)	2.84
Shannon index, median (Control)	2.28
Mean Shannon difference (ALS minus Control)	+0.59
Paired Wilcoxon signed-rank statistic	0.0
Paired Wilcoxon p-value	0.004
PERMANOVA pseudo-F (Bray-Curtis)	1.71
PERMANOVA p-value (999 permutations)	0.025
Top genus by adjusted p-value	*Parabacteroides* (q = 0.068)
Top 5 genera (by q)	*Parabacteroides*, *Ruminococcus torques* group, *Blautia*, *Subdoligranulum*, *Lachnoclostridium*

4 Discussion

The clearest result is that ALS patients had higher Shannon diversity than their spouses ($p = 0.004$), the opposite of what I had predicted. There are two ways to read this. One is that my prediction was simply wrong: the literature on ALS alpha diversity is not consistent, with some studies reporting lower diversity, others no difference or an increase (Blacher et al. 2019). The other reading is that a high Shannon value is not always a good sign. If a dominant beneficial taxon is lost, the remaining taxa become more evenly distributed and the Shannon index goes up even when the community as a whole may be in a worse state. Distinguishing between the two readings would need longitudinal sampling, which this dataset does not include.

The PERMANOVA result (pseudo-F = 1.7, p approximately 0.02) means that the two groups differ in overall composition. The effect is modest, however, and the PCoA does not show clear clustering. With only 18 samples, a borderline result like this should be interpreted with caution. The visual overlap in the PCoA is itself informative: two groups can be statistically different and still sit in similar regions of ordination space, especially when the test is picking up many small shifts rather than one large effect.

Several of the top genera have been linked to gut inflammation or neurodegenerative disease in previous work. The top hit, *Parabacteroides* (mean 2.0 percent in ALS versus 0.04 percent in spouses, q approximately 0.07), has appeared in earlier Parkinson's and ALS surveys, although the direction of the reported shift is not consistent across studies. The *Ruminococcus torques* group, also enriched in ALS, is a mucin-degrading taxon associated with intestinal barrier dysfunction and inflammatory bowel disease; barrier dysfunction is the main route through which gut bacteria engage the immune system, the second of the three gut-brain axis channels described in the Introduction. *Blautia* and *Subdoligranulum*, both short-chain fatty acid producers usually considered beneficial, were also higher in ALS samples. Their main hypothesized effect on the brain is through the metabolite route, the third gut-brain axis channel (Cryan et al. 2019). Enrichment of SCFA producers in the ALS group is initially counter-intuitive, but it is consistent with the elevated Shannon diversity: the ALS samples appear to contain more of several genera at moderate abundance rather than a clear depletion of any single one. The only top hit enriched in spouses was *Escherichia-Shigella*, a Proteobacteria group that contains many pathobionts; its higher abundance in the controls most likely reflects individual variation at this sample size. *Akkermansia muciniphila*, the species whose supplementation slowed disease progression in the SOD1 mouse model (Blacher et al. 2019), did not appear among the top hits in either direction in this cohort.

The paired design is the main strength of the Hertzberg cohort and of this reanalysis. Spouses share diet, household, and many environmental exposures, which are precisely the variables that confound case-control comparisons between unrelated individuals. By comparing within pairs, much of that variation was removed from the analysis, and this is the reason the paired Wilcoxon test reaches significance with only nine pairs. The trade-off is that the comparison is between an ALS patient and their cohabiting spouse, not between an ALS patient and a randomly selected healthy adult, so the result describes the local within-pair difference rather than a population-level difference.

Several limitations should be kept in mind when interpreting these results. The main one is the sample size: a single-cohort study with 18 samples is preliminary, and replication in independent cohorts is needed before any of these findings can be considered established. The 16S V4 region resolves taxonomy reliably only to the genus level, so species-level claims are not supported and functional questions would need shotgun metagenomics. The cohort was recruited from a single geographic region, which limits generalizability to ALS patients with different diets or baseline microbiomes. The prevalence filter, although standard, removes rare taxa that may carry real biological signal, and the 5 percent threshold is a defensible but arbitrary choice. The per-genus tests are also underpowered, and the FDR correction controls only the expected fraction of false positives among the called genera, not the truth of any individual hit. Even with these limitations, the reanalysis adds another data point to the question of whether the gut microbiome is altered in ALS. The paired Wilcoxon and the PERMANOVA both detected real differences between ALS patients and their spouses, although the direction of the alpha-diversity effect was the opposite of what I had predicted. The most useful next step would be a meta-analysis combining this cohort with other public ALS 16S datasets, ideally followed by shotgun metagenomic work so that functional pathways can be examined directly. Replication across many cohorts is what will decide whether the gut microbiome is part of the ALS disease process or only a downstream consequence of it.

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